

TUMOR DEVELOPMENT MODEL V2

Tumor phase-field

$\phi = 0$ medium

$\phi = 1$ tumor

in code: $\phi_{max} = 0.95$

$$\frac{\partial \phi}{\partial t} = -m [0.5(1-\phi)\phi(1-2\phi) - \epsilon \nabla^2 \phi] + G\phi(1-\phi)$$

mobility

reaction

surface tension

growth

• SURFACE TENSION

$\epsilon^2 \rightarrow$ interface width

$\hookrightarrow$ generalized Laplacian for variable interface width
[if ϵ constant $\rightarrow$ normal Laplacian]

$\hookrightarrow$ growth dependent on phenotype

$\Rightarrow$ we make ϵ^2 dependent on space

$$\epsilon_{xy} = \epsilon^2 (1 + \alpha_\epsilon \cdot m)$$

$\hookrightarrow$ we clip it in code to avoid numerical instability

$m = u_{tot} / n_m$ normalized u , n_m - normalization constant (0.4)

m = local coupling metric

$\hookrightarrow \alpha_\epsilon$ how strongly phenotype field modifies phase-field interfacial width (baseline)

$\Rightarrow$ Replace $\epsilon^2 \nabla^2 \phi$ by $\nabla(\epsilon_{xy} \nabla \phi)$

[divergence discretization] $\hookrightarrow$ Cahn-Hilliard

$\hookrightarrow$ average ϵ^2 between neighbours, taking finite differences in each direction

• GROWTH

local growth weighted by nutrients + phenotypes

$$G = \sum_{k \in p} r_k \cdot \frac{c}{c + K_c} \cdot \frac{u_k}{u_{tot}}$$

r_k = proliferation rate

$c(x,y)$ = nutrient concentration

K_c = half saturation rate

u_k = concentration cells of the phenotype

$$\frac{G\phi(1-\phi)}{\text{only in tumor}}$$

Nutrient diffusion - consumption

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c - \lambda \phi c$$

diffusion

relaxation to background

λ = external replenishment

$\phi \rightarrow$ only tumor consumes

$$- \gamma_c \phi \sum_{k \in p} u_k q_k$$

uptake

q_k = nutrient consumption rate

γ_c = global strength nutrient consumption

Phenotype densities (phenotype p)

$$U_{tot} = \sum_{k \in P} U_k$$

↳ population dynamics

$$\frac{\partial U_k}{\partial t} = \underbrace{D_k \nabla^2 U_k}_{\text{diffusion}} + \underbrace{r_k \frac{c}{c+K_c} U_k C_k}_{\substack{\text{monod nutrient dependence} \\ \text{competition based} \\ \text{proliferation} \\ \text{(growth)}}} - \underbrace{d_k U_k}_{\text{death}}$$

• PROLIFERATION (GROWTH)

r_k = max. proliferation rate

K_c = half saturation content
($r_k/2$)

C_k = competition factor

$r_k \frac{c}{c+K_c}$

nutrient
limited proliferation

$U_k C_k$

competition
phenotypes

$$\hookrightarrow E_k = 1 - \frac{U_{tot}}{K_u}$$

K_u = carrying capacity
of total U

• DEATH

d_k = death rate

U_k = concentration cells phenotype k

$-d_k U_k$